Transport Booking System

Using Queue and Linked List in C++

- Student Name : M.Moiz Ur Rehman
- Roll no : 24010101-085
- Course: Data Structures
- Instructor : Ma’am Azka Mir
- Date:18-03-2026

Introduction

A console-based C++ application designed to manage transport bookings efficiently through automated data structures.

Core Functions

- Store vehicle information
- Handle booking requests
- Confirm bookings

Data Structures

- Queue for pending requests
- Linked List for confirmed bookings

Problem & Objectives

Problem

Manual booking systems are slow and inefficient. Managing requests and confirmations is difficult.

Objectives

- Automate booking system
- Implement Queue (FIFO)
- Implement Linked List
- Improve data handling efficiency

System Overview

Three main modules work together to manage transport bookings efficiently.

01

Vehicle Management

Add and view vehicles in the system

02

Request Handling

Store booking requests in a queue

03

Booking Confirmation

Store confirmed bookings in linked list

Data Structures Used

Queue (FIFO)

- Stores booking requests
- First request is processed first
- Efficient request management

Linked List

- Stores confirmed bookings
- Dynamic memory allocation
- Easy insertion of new data

Linked List Structure

Confirmed Bookings

Each node contains booking details:

- Booking ID
- Customer Name
- Vehicle Type
- Source
- Destination

append()

Add new booking

display()

Show all bookings

Queue Implementation

Pending Requests

Implemented using arrays with circular queue logic for efficiency.

enqueue()

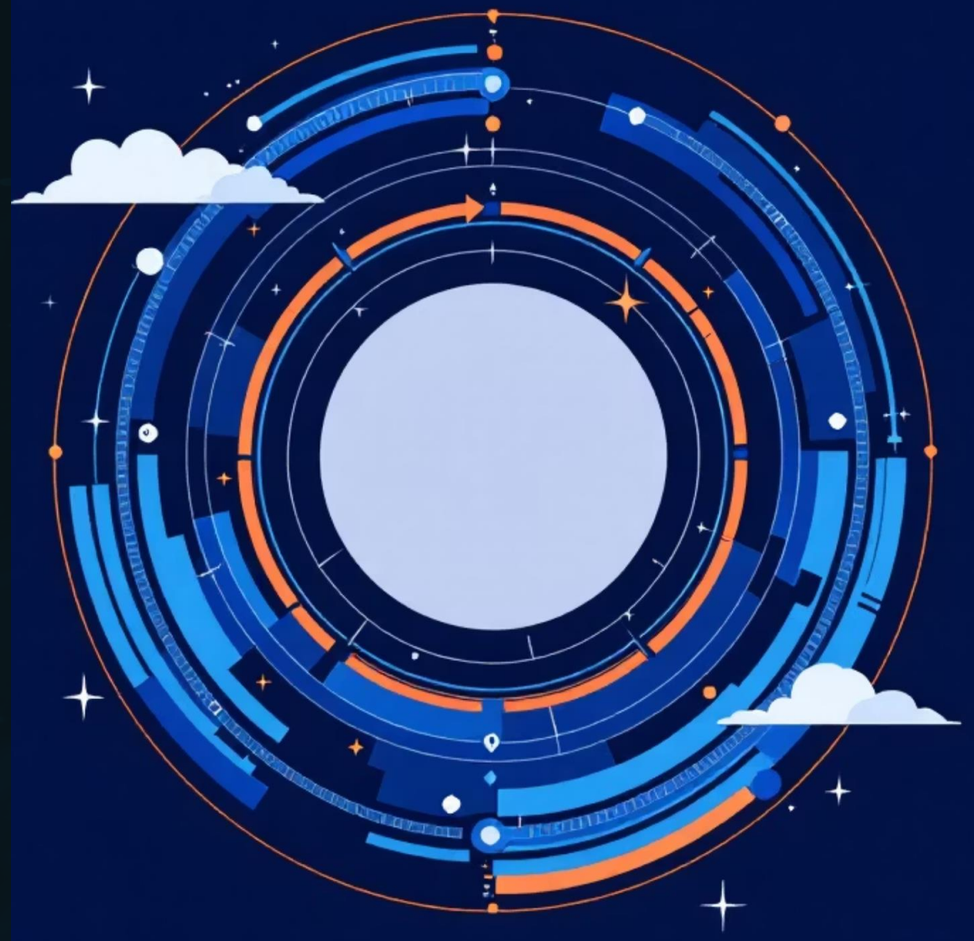
Add request to queue

dequeue()

Process request

print()

View pending requests



Vehicle Management

Stored Data

- vehicleID
- vehicleName
- vehicleCapacity

Features

- Add vehicle (max 5)
- View all vehicles

Working Flow

- User enters booking request
- Request added to queue
- System processes request
- Request removed from queue
- Booking stored in linked list

Menu Options

- Add Vehicle
- View Vehicles
- Request Booking
- Process Booking
- View Confirmed
- View Pending

Conclusion

System Success

- Queue for request handling
- Linked List for confirmed bookings

Advantages

- Efficient and organized
- Easy to understand
- Real-world application

Future Improvements

- Database integration
- GUI interface
- Unlimited storage